Appendix S: Falsifiable Predictions

Testable Hypotheses Derived from the (C, Ψ, C^*, K, Q) Framework

For: Complementarity and Context: A Unified Framework for Economic Rationality

Framework Validation Document

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Abstract

A framework that can explain everything explains nothing. This appendix addresses the Popperian challenge by deriving **specific, falsifiable predictions** from the (C, Ψ, C^*, K, Q) framework. We identify 10 testable hypotheses, specify their falsification criteria, and outline empirical strategies for testing them. The goal is to transform the framework from an organizing schema into a productive scientific research program that generates surprising, testable predictions—the standard established by Becker, Stiglitz, and other foundational economic frameworks.

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1 The Falsifiability Challenge

1.1 The Problem

The (C, Ψ, C^*, K, Q) framework claims that:

1. Complementarities (C) exist between economic inputs
2. Context (Ψ) determines which complementarities are active
3. The effective complementarity structure $C^*(\Psi)$ varies systematically
4. Knowledge asymmetries (K) affect outcomes
5. All of this determines observable outcomes (Q)

The critique: If any empirical finding can be reinterpreted as "a complementarity moderated by context," the framework is unfalsifiable and therefore unscientific.

1.2 The Becker/Stiglitz Standard

Becker's rational choice framework and Stiglitz's information economics faced the same critique. They survived because they generated:

1. **Quantitative predictions**: Not just "crime responds to incentives" but $\frac{\partial Crime}{\partial p} < 0$ with estimable elasticities
2. **Surprising implications**: Predictions that were not obvious before the theory
3. **Institutional specificity**: Which institutions arise under which conditions
4. **Ex-ante testability**: Predictions made before observing outcomes

1.3 Our Strategy

We derive 10 predictions that are:

- **Specific**: Clear about magnitudes and directions
- **Surprising**: Not obvious without the framework
- **Falsifiable**: We specify what evidence would refute them
- **Testable**: Data exists or can be collected

2 Prediction 1: Cash Transfer Effectiveness and Social Capital

Prediction 1 (Cash Transfers and Social Capital)**.** *Unconditional cash transfers (UCTs) will show* ***smaller*** *treatment effects in communities with high pre-existing social capital, because informal insurance mechanisms already address the constraints that cash relaxes.*

2.1 Framework Derivation

The framework posits:

$$Q = f(I, \Psi) \quad \text{where} \quad \frac{\partial^2 Q}{\partial I \partial \Psi_{social}} < 0 \tag{1}$$

Cash (I) and social capital (Ψ_{social}) are **substitutes**, not complements, for consumption smoothing. This is surprising because we typically expect social capital to enhance interventions.

2.2 Formal Hypothesis

Hypothesis 1. *Let $\tau(\Psi_s)$ be the average treatment effect of UCTs in communities with social capital index Ψ_s. Then:*

$$\frac{d\tau}{d\Psi_s} < 0 \tag{2}$$

Falsification Criterion

The prediction is **falsified** if:

- UCT effects are larger in high social capital communities ($\frac{d\tau}{d\Psi_s} > 0$), OR
- UCT effects are invariant to social capital ($\frac{d\tau}{d\Psi_s} = 0$)

Empirical Test

Data: GiveDirectly RCTs in Kenya, Uganda, Rwanda with village-level social capital measures (trust surveys, group membership, informal lending networks).
Method: Estimate $\tau_i = \alpha + \beta \cdot \Psi_{s,i} + X_i'\gamma + \varepsilon_i$ using village-level treatment effects.
Test: $H_0 : \beta \geq 0$ vs. $H_1 : \beta < 0$

2.3 Why This Is Surprising

Standard intuition: Social capital is good, cash is good, together they should be better. The framework predicts the opposite—they substitute for the same function (insurance against shocks).

3 Prediction 2: Management Practices and Trust

Prediction 2 (Management and Trust Complementarity)**.** *The productivity gains from adopting "high-performance" management practices will be **larger** in high-trust societies than in low-trust societies, because these practices require employee autonomy and cooperation.*

3.1 Framework Derivation

From Bloom's research, management practices (M) increase productivity. From the framework:

$$C^{M,productivity}(\Psi_{trust}) \quad \text{where} \quad \frac{\partial C^{M,Q}}{\partial \Psi_{trust}} > 0 \tag{3}$$

Management practices and trust are **complements**—each raises the return to the other.

3.2 Formal Hypothesis

Hypothesis 2. *Let $\beta_M(\Psi_t)$ be the productivity elasticity with respect to management in countries with trust level Ψ_t. Then:*

$$\frac{d\beta_M}{d\Psi_t} > 0 \tag{4}$$

Falsification Criterion

The prediction is **falsified** if:

- Management practices are equally effective regardless of trust ($\frac{d\beta_M}{d\Psi_t} = 0$), OR
- Management practices are *more* effective in low-trust societies ($\frac{d\beta_M}{d\Psi_t} < 0$)

Empirical Test

Data: World Management Survey (WMS) + World Values Survey trust measures + firm productivity data across 35+ countries.
Method: Estimate $\ln Q_{ijc} = \alpha + \beta_1 M_{ijc} + \beta_2 Trust_c + \beta_3(M \times Trust) + X'\gamma + \varepsilon$
Test: $H_0 : \beta_3 \leq 0$ vs. $H_1 : \beta_3 > 0$

3.3 Quantitative Prediction

Based on Bloom et al. (2012) on decentralization and trust, we predict:

$$\beta_3 \approx 0.02 - 0.04 \tag{5}$$

A 10-point increase in trust (0-100 scale) should increase the management-productivity elasticity by 0.2-0.4 percentage points.

4 Prediction 3: Early Childhood Programs and Baseline Disadvantage

Prediction 3 (Heckman Curve Heterogeneity)**.** *The rate of return to early childhood investment will be **higher** for more disadvantaged children, but the rate of decline with age will also be **steeper** for disadvantaged children.*

4.1 Framework Derivation

From Heckman's technology of skill formation:

$$\theta_{t+1} = f(\theta_t, I_t) \quad \text{with} \quad \frac{\partial^2 f}{\partial \theta_t \partial I_t} > 0 \tag{6}$$

Dynamic complementarity implies that low initial θ_0 (disadvantage) means:

1. Higher marginal return to early investment (more room to grow)
2. Faster decline of returns with age (without early investment, the gap compounds)

4.2 Formal Hypothesis

Hypothesis 3. *Let $r(t, \theta_0)$ be the rate of return to investment at age t for children with initial endowment θ_0. Then:*

$$\left.\frac{\partial r}{\partial \theta_0}\right|_{t=0} < 0 \quad \textit{(higher returns for disadvantaged at early ages)} \tag{7}$$

$$\frac{\partial^2 r}{\partial t \partial \theta_0} > 0 \quad \textit{(steeper decline for disadvantaged)} \tag{8}$$

Falsification Criterion

The prediction is **falsified** if:

- Returns decline at the same rate regardless of baseline disadvantage
- Advantaged children show steeper decline (opposite pattern)

Empirical Test

Data: Perry Preschool, Abecedarian, Head Start Impact Study with baseline SES measures and age-varying intervention points.
Method: Meta-regression of treatment effects on (age at intervention × baseline SES) interaction.
Test: Interaction coefficient > 0 indicates steeper decline for disadvantaged.

5 Prediction 4: Trade Shocks and Labor Market Institutions

Prediction 4 (China Shock Heterogeneity)**.** *The negative employment effects of Chinese import competition will be **smaller** in countries/regions with flexible labor markets and active retraining programs than in those with rigid labor markets.*

5.1 Framework Derivation

From Autor (China Shock) and Bloom (Trade and Innovation):

$$\Delta Employment = \beta \cdot \Delta Imports + \gamma \cdot (\Delta Imports \times \Psi_{labor\ market}) + \varepsilon \tag{9}$$

The framework predicts $\gamma > 0$: flexible labor markets (Ψ_{flex} high) enable reallocation that offsets displacement.

5.2 Formal Hypothesis

Hypothesis 4. *Let β_{EU} be the employment effect of import competition in Europe (flexible, with retraining) and β_{US} be the effect in the US (less flexible, limited retraining). Then:*

$$|\beta_{EU}| < |\beta_{US}| \tag{10}$$

Falsification Criterion

The prediction is **falsified** if:

- Employment effects are larger in flexible labor markets
- Employment effects are identical across institutional contexts

Empirical Test

Data: Autor et al. (2013) for US; Bloom et al. (2016) for Europe; harmonized import exposure measures.
Method: Compare estimated β coefficients across institutional contexts with same identification strategy.
Preliminary Evidence: Bloom et al. find *positive* innovation effects in Europe where Autor et al. find negative employment effects in US. This is consistent with the prediction but needs direct comparison.

6 Prediction 5: Microfinance and Entrepreneurial Ability

Prediction 5 (Microfinance Heterogeneity)**.** *Microfinance will show **positive** effects only for borrowers in the top quartile of baseline entrepreneurial ability, and **zero or negative** effects for the bottom quartile.*

6.1 Framework Derivation

From Duflo's microfinance research:

$$C^{credit,ability} > 0 \tag{11}$$

Credit and ability are complements. For those with low ability, credit may fund consumption or unprofitable ventures.

6.2 Formal Hypothesis

Hypothesis 5. *Let $\tau(q)$ be the treatment effect of microfinance for borrowers in ability quartile $q \in \{1, 2, 3, 4\}$. Then:*

$$\tau(4) > 0 \quad \textit{(top quartile benefits)} \tag{12}$$

$$\tau(1) \leq 0 \quad \textit{(bottom quartile does not benefit)} \tag{13}$$

$$\tau(4) - \tau(1) > 0 \quad \textit{(heterogeneity)} \tag{14}$$

Falsification Criterion

The prediction is **falsified** if:

- Microfinance helps low-ability borrowers as much as high-ability
- Treatment effect heterogeneity is not related to baseline ability

Empirical Test

Data: Banerjee et al. (2015) Hyderabad RCT with baseline business characteristics; Banerjee et al. (2019) long-run follow-up.
Method: Quantile treatment effects or machine learning (causal forests) to identify heterogeneity by baseline characteristics.
Preliminary Evidence: Banerjee et al. (2019) find exactly this pattern—"gung-ho entrepreneurs" benefit, others do not.

7 Prediction 6: Information Interventions and Prior Beliefs

Prediction 6 (Bayesian Updating Asymmetry)**.** *Information interventions will be* ***more effective*** *when they contradict weak priors than when they contradict strong priors, because Bayesian updating is proportional to prior uncertainty.*

7.1 Framework Derivation

Knowledge (K) affects outcomes, but the effect depends on prior beliefs ($\Psi_{beliefs}$):

$$\Delta Behavior = f(\Delta K, \sigma^2_{prior}) \tag{15}$$

where $\frac{\partial f}{\partial \sigma^2_{prior}} > 0$—information matters more when priors are diffuse.

7.2 Formal Hypothesis

Hypothesis 6. *Let $\tau_{info}(\sigma^2)$ be the treatment effect of information provision for individuals with prior variance σ^2. Then:*

$$\frac{d\tau_{info}}{d\sigma^2} > 0 \tag{16}$$

Falsification Criterion

The prediction is **falsified** if:

- Information effects are larger for those with strong priors (confident beliefs)
- Information effects are invariant to prior confidence

Empirical Test

Data: Any RCT with information treatment + elicited prior beliefs (e.g., Dupas 2011 on HIV risk; Jensen 2010 on returns to education).
Method: Estimate $\tau_i = \alpha + \beta \cdot \sigma^2_{prior,i} + X_i'\gamma + \varepsilon_i$
Test: $H_0 : \beta \leq 0$ vs. $H_1 : \beta > 0$

8 Prediction 7: Graduation Programs and Component Necessity

Prediction 7 (Bundle Decomposition)**.** *Removing **any single component** from the Graduation bundle (asset, training, coaching, savings, health) will reduce program effectiveness by more than that component's standalone effect—demonstrating superadditivity.*

8.1 Framework Derivation

The Graduation approach works because components are complements:

$$f(A,T,C,S,H) > f(A,0,0,0,0)+f(0,T,0,0,0)+f(0,0,C,0,0)+f(0,0,0,S,0)+f(0,0,0,0,H) \tag{17}$$

This implies that removing any component destroys more value than just that component's marginal contribution.

8.2 Formal Hypothesis

Hypothesis 7. *Let τ_{full} be the effect of the full Graduation bundle and τ_{-j} be the effect with component j removed. Then:*

$$\tau_{full} - \tau_{-j} > \tau_j \quad \forall j \tag{18}$$

where τ_j is the standalone effect of component j.

Falsification Criterion

The prediction is **falsified** if:

- Removing a component reduces effectiveness by exactly that component's standalone value (additivity)
- Removing a component has no effect (redundancy)

Empirical Test

Design: Factorial RCT with all $2^5 = 32$ combinations of Graduation components.
Challenge: This is expensive. Alternative: Compare existing Graduation RCTs that vary in implementation (some sites skipped components due to logistics).
Preliminary Evidence: Banerjee et al. (2015) did not test this directly, but site-level variation in implementation intensity correlates with effectiveness.

9 Prediction 8: Uncertainty and Investment Responsiveness

Prediction 8 (Uncertainty Dampening)**.** *The elasticity of firm investment with respect to Tobin's Q will be* ***lower*** *during high-uncertainty periods than during low-uncertainty periods.*

9.1 Framework Derivation

From Bloom's uncertainty research:

$$\frac{\partial I}{\partial Q} = g(\sigma^2) \quad \text{where} \quad g'(\sigma^2) < 0 \tag{19}$$

Uncertainty creates real options value of waiting, reducing responsiveness to fundamentals.

9.2 Formal Hypothesis

Hypothesis 8. *Let β_Q^{high} and β_Q^{low} be the investment-Q elasticities during high and low VIX periods. Then:*

$$\beta_Q^{high} < \beta_Q^{low} \tag{20}$$

Falsification Criterion

The prediction is **falsified** if:

- Investment is equally responsive to Q regardless of uncertainty
- Investment is *more* responsive during uncertain times

Empirical Test

Data: Compustat firm-level data + VIX or Baker-Bloom-Davis EPU index.
Method: Estimate $I_{it} = \alpha + \beta_1 Q_{it} + \beta_2(Q \times High_VIX_t) + X'\gamma + \varepsilon$
Test: $H_0 : \beta_2 \geq 0$ vs. $H_1 : \beta_2 < 0$
Preliminary Evidence: Bloom et al. (2007) find exactly this pattern in UK data.

10 Prediction 9: Female Leadership and Policy Domains

Prediction 9 (Representation and Responsiveness)**.** *Female political leaders will increase public goods provision in domains that disproportionately affect women (water, health, childcare) but* ***not*** *in domains with gender-neutral preferences (roads, electricity in non-water-carrying contexts).*

10.1 Framework Derivation

From Duflo's political economy research:

$$C^{female\ leader, female-preferred\ goods} > 0 \tag{21}$$

but

$$C^{female\ leader, gender-neutral\ goods} \approx 0 \tag{22}$$

10.2 Formal Hypothesis

Hypothesis 9. *Let τ_j be the effect of female leadership on public good j. Then:*

$$\tau_{water} > 0, \quad \tau_{health} > 0, \quad \tau_{childcare} > 0 \tag{23}$$

$$\tau_{roads} \approx 0, \quad \tau_{electricity} \approx 0 \tag{24}$$

Falsification Criterion

The prediction is **falsified** if:

- Female leaders increase all public goods equally (no domain specificity)
- Female leaders increase gender-neutral goods more than female-preferred goods

Empirical Test

Data: Chattopadhyay & Duflo (2004) Panchayat data; replications in other Indian states; extensions to other countries with gender quotas.
Method: Compare treatment effects across public good categories.
Preliminary Evidence: Original findings support this pattern—water investment increased, but domain-specificity not fully tested.

11 Prediction 10: Skill Complementarity and Automation

Prediction 10 (Automation and Skill Bundles)**.** *Automation will **increase** demand for workers who combine technical skills with social/creative skills, while **decreasing** demand for workers with only technical skills or only social skills.*

11.1 Framework Derivation

From Autor's task framework and Heckman's skill complementarities:

$$C^{technical, social}(\Psi_{automation}) \quad \text{where} \quad \frac{\partial C}{\partial \Psi_{auto}} > 0 \tag{25}$$

Automation raises the complementarity between skill types—neither alone is sufficient, but together they are more valuable.

11.2 Formal Hypothesis

Hypothesis 10. *Let $w(T, S, A)$ be wages as a function of technical skills T, social skills S, and automation exposure A. Then:*

$$\frac{\partial^3 w}{\partial T \partial S \partial A} > 0 \tag{26}$$

The interaction between technical and social skills becomes more positive as automation increases.

Falsification Criterion

The prediction is **falsified** if:

- Automation increases returns to technical skills alone
- Automation increases returns to social skills alone
- The T-S complementarity is unchanged by automation

Empirical Test

Data: O*NET skill requirements + occupation-level automation exposure (Frey & Osborne; Webb) + wage data (ACS, CPS).
Method: Estimate wage equations with three-way interaction (Technical × Social × Automation).
Test: Coefficient on three-way interaction > 0.
Preliminary Evidence: Deming (2017) shows rising returns to social skills; Autor & Dorn show polarization; the three-way interaction is not yet tested.

12 Summary: The Falsification Matrix

#	Prediction	Framework Element	Surprising?	Data Exists?
1	Cash transfers × social capital	$C^{I,\Psi} < 0$ (substitutes)	Yes	Yes
2	Management × trust	$C^{M,Q}(\Psi_{trust})$	Moderate	Yes
3	Heckman curve heterogeneity	Dynamic $C(\theta_0)$	Yes	Partial
4	Trade shock × labor institutions	$C^{shock,\Psi_{flex}}$	Moderate	Yes
5	Microfinance × ability	$C^{credit,ability}$	Moderate	Yes
6	Information × prior beliefs	$K \times \Psi_{beliefs}$	Yes	Partial
7	Graduation bundle decomposition	Superadditivity	Yes	No (needs RCT)
8	Uncertainty × investment	$\sigma^2 \to C^{Q,I}$	Moderate	Yes
9	Female leaders × policy domain	$C^{gender,domain}$	Moderate	Yes
10	Automation × skill bundles	$C^{T,S}(\Psi_{auto})$	Yes	Partial

Table 1: Summary of 10 falsifiable predictions

13 What Would Falsify the Entire Framework?

Individual predictions can fail while the framework survives (any specific C^{ij} might be mis-specified). But the **entire framework** would be falsified by:

13.1 Strong Falsification

1. **Context invariance**: If complementarities are identical across all institutional contexts ($C^*(\Psi_1) = C^*(\Psi_2)$ for all Ψ_1, Ψ_2), then the Ψ-dependence is empty.

2. **Strict additivity**: If all production functions are additive ($f(x, y) = g(x) + h(y)$ always), then complementarities don't exist.

3. **Information irrelevance**: If K-interventions never affect outcomes, then knowledge is not a framework element.

13.2 Weak Falsification

1. **Unpredictable heterogeneity**: If treatment effect heterogeneity cannot be predicted by observable Ψ, then the framework has no predictive power (even if true).

2. **Wrong signs**: If most predictions have wrong signs (e.g., complements turn out to be substitutes), then our understanding of the C-matrix is incorrect.

13.3 The Meta-Test

If 7 or more of the 10 predictions are confirmed, the framework is productive.

If 5 or more are falsified, the framework needs fundamental revision.

If predictions are confirmed but not surprising, the framework is correct but not useful.

14 Research Agenda

14.1 Immediate Tests (Data Exists)

1. **Prediction 2** (Management $\times$ Trust): Re-analyze WMS data with trust interactions

2. **Prediction 5** (Microfinance $\times$ Ability): Re-analyze Banerjee et al. with ability quartiles

3. **Prediction 8** (Uncertainty $\times$ Investment): Replicate Bloom et al. with updated data

14.2 Medium-Term Tests (Requires New Analysis)

1. **Prediction 1** (Cash × Social Capital): Analyze GiveDirectly data with village-level social capital
2. **Prediction 4** (Trade × Institutions): Harmonize Autor and Bloom datasets
3. **Prediction 10** (Automation × Skills): Construct three-way interaction estimates

14.3 Long-Term Tests (Requires New Data Collection)

1. **Prediction 7** (Graduation Decomposition): Factorial RCT with component variation
2. **Prediction 6** (Information × Priors): RCT with elicited prior beliefs

15 Conclusion

This appendix transforms the (C, Ψ, C^*, K, Q) framework from an organizing schema into a **testable scientific theory**. We have:

1. Derived **10 specific, falsifiable predictions**
2. Specified **what evidence would refute each prediction**
3. Identified **data sources and empirical strategies**
4. Defined **what would falsify the entire framework**

The framework now meets the Becker/Stiglitz standard: it generates surprising predictions with specified falsification criteria. Whether these predictions are confirmed or refuted, the framework will have contributed to cumulative scientific knowledge.

The ultimate test: Can we predict, *before seeing the data*, which interventions will work in which contexts? If yes, the framework is scientifically productive. If no, it remains a useful organizing device but not a predictive theory.